Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

ML engineer with a **full applied ML lifecycle** - built AgentProbe, an open-source system with LLM-powered evaluation, RAG retrieval pipelines, query decomposition agents, and automated prompt iteration. 8+ years production backend engineering (1M+ records, 100K+ MAU, Python/AWS). Open-source contributor. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LLM Evaluation & Judges | RAG Pipelines & Retrieval | Prompt Engineering & Iteration | Agentic Query Decomposition | ML Lifecycle (Data → Prod) | Feature Engineering | LangChain / LangGraph | ChromaDB / Embeddings | Python / FastAPI | NLP / Intent Modeling | Open-Source ML | AWS / Production Systems

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **distributed data pipelines** on AWS (S3, Glue, EC2) processing **1M+ records** with automated quality checks - demonstrating the production systems discipline needed to move ML from experiment to reliable infrastructure.
- Improved production query performance by **30%** by architecting PostgreSQL schemas, data access layers, and indexing strategies across 20+ projects - the same analytical rigor applied to offline metric improvement.
- Reduced deployment times by **40%** and maintained sub-10 bug count for 6+ consecutive months via TDD/BDD gates across **120+ Agile sprints** for **100K+ monthly active users**.
- Cut microservices execution time by **50%** via Docker-based containerization; achieved **99.5% uptime** across multi-tenant production environments.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: Applied ML Evaluation System OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure

Built a **closed-loop applied ML system**: query decomposition (Orchestrator agent routes and decomposes queries into sub-tasks), **RAG retrieval evaluation** (RAG Tracer scores retrieval quality and chunk relevance per query), **LLM-powered judges** (5-metric engine: correctness, completeness, hallucination, retrieval ranking, compliance), and **Auto Fix agent** (automated prompt iteration triggered when eval scores degrade). 15-endpoint FastAPI backend, React observability dashboard, full trace visibility. Supports Ollama, OpenAI, Anthropic, Groq - 100% local at zero cost. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP Query Understanding for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Built **query understanding pipeline** mapping natural language intent (EN/ES) to structured FHIR API calls via LLMs, BERT, and spaCy. Intent modeling, multilingual query rewriting, HIPAA-compliant RBAC. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle on UK road accident data: feature engineering, model development (Random Forest + Logistic Regression), rigorous evaluation, and deployed interactive dashboard. **85%+ accuracy**; outputs designed for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & ML: LangChain, LangGraph, LLM Judges, RAG, ChromaDB, Embeddings, Prompt Engineering, Iterative Model Improvement, Ollama, OpenAI, Anthropic, Groq, FastAPI

ML Stack: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering, Predictive Modeling

Databases: PostgreSQL, MySQL, Oracle, NoSQL, GraphQL - schema design, query tuning, indexing

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

Testing: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum